

## Lab 11 (Phishing Analysis Fundamentals)

### exercise 1

Domain:

A helpful analogy is to think of an email address as a home mailing address:

- The **domain** is like the street or apartment building
- The **username** is the specific person or mailbox within that location

With both pieces of information, the postal worker (mail server) knows exactly where to deliver the message. In the next task, we'll explore the network protocols responsible for sending and receiving emails.

Answer the questions below

Identify the domain used in the following email address:

hatsalesman@tryhatme.com

tryhatme.com

✓ Correct Answer

- **Simple Mail Transfer Protocol (SMTP): Sends emails**
- **Post Office Protocol (POP3): Downloads emails to a device**
- **Internet Message Access Protocol (IMAP): Syncs emails across devices**

Which protocol is responsible for sending an email from a client to a mail server?

SMTP

✓ Correct Answer

Which service is used to look up the recipient domain's mail server?

DNS

✓ Correct Answer

Bob wants to access his email from multiple devices, including his phone and laptop. Which protocol should he use?

IMAP

✓ Correct Answer

**Subject:** Help protect your budget by protecting your home

**X-ORIGINATING IP:** 43.255.56.161

From ADT Security Services <newsletters@ant.anki-tech.com> @  
To alexa@yahoo.com @  
Reply to reply@ant.anki-tech.com @  
Subject **Help protect your budget by protecting your home**

Source of: file:///home/ubuntu/Desktop/Email%20Samples/email1.eml — □ ×

File Edit View Help

Received: from 10.222.142.150  
by atlas206.free.mail.ne1.yahoo.com with HTTPS; Mon, 21 Jun 2021 15:36:02 +0000  
Return-Path: <reback-a3970-837890-838253-c8b776d9=952622232=8@ant.anki-tech.com>  
X-Originating-Ip: [43.255.56.161]  
Received-SPF: pass (domain of ant.anki-tech.com designates 43.255.56.161 as permit  
Authentication-Results: atlas206.free.mail.ne1.yahoo.com;  
dkim=pass header.i=@ant.anki-tech.com header.s=default;  
spf=pass smtp.mailfrom=ant.anki-tech.com;  
dmarc=pass(p=NONE) header.from=ant.anki-tech.com;  
X-Apparently-To: alexa@yahoo.com; Mon, 21 Jun 2021 15:36:02 +0000  
X-YMailISG: iU.RbH8WLDuS8PKXbPwmeJ5ksCZTcrgGSzQNPLV2GG3TS1LJ  
tLtofc0wExjmLmWTFhcEr1guoWTIy09uPLS1g2sv9ZNXf366atDDf8yKQApo  
rfdxFKAERJalk4hzdsHGAinSPoQR6AZmaFo83Hso0emd80z7hjSYwHAjfpZn  
G9EYqjGm8Krb5Wf9RVTqVUh\_xam0JSRA7Srl1b3d73aea31ilE0b1ddFzZ\_W  
Wl37yrp9KU6\_dIWfGR.1pABp95cRj\_mDJUUpvJnSpMfer0r8Jj70BJ03VAdnx

What is the full subject line of `email1.eml` ?

Help protect your budget by protecting your home

✓ Correct Answer

View the message source of `email1.eml` using Thunderbird in your VM.  
What the IP address listed as the `X-Originating-Ip` ?

43.255.56.161

✓ Correct Answer

**Filename: zmqpalgh.pdf**



THM{BENIGN\_PDF\_ATTACHMENT}

Answer the questions below

Open up the `email12.txt` file to view the source of an attachment.  
What is the `Content-Type` of the attachment?

`application/pdf`

✓ Correct Answer

What is the name of the attachment from the previous question?

`zmqpqlgh.pdf`

✓ Correct Answer

Decode the base64 string using either a [PDF converter \(opens in new tab\)](#) or [CyberChef \(opens in new tab\)](#).  
What is the hidden flag value?

`THM{BENIGN_PDF_ATTACHMENT}`

✓ Correct Answer

## Email3.eml:

From Thank you! Home Depot <support@teckbe.com> @

Source of: file:///home/ubuntu/Desktop/Email%20Samples/email3.eml

File Edit View Help

```
Received: from 10.253.62.157
  by atlas102.free.mail.gql.yahoo.com with HTTPS; Sun, 11 Jul 2021 11:48:13 +0000
Return-Path: <support@teckbe.com>
X-Originating-Ip: [103.234.236.83]
Received-SPF: pass (domain of teckbe.com designates 103.234.236.83 as permitted sender)
Authentication-Results: atlas102.free.mail.gql.yahoo.com;
  dkim=pass header.i=@teckbe.com header.s=dk2048;
  spf=pass smtp.mailfrom=teckbe.com;
  dmarc=pass (p=NONE) header.from=teckbe.com;
X-Apparently-To: alexa@yahoo.com; Sun, 11 Jul 2021 11:48:13 +0000
X-YMailISG: Pcp93.8WLDsI7m7VAEQNTYkM0dxkhTUX7phGwD48dol2XbE6
  Gitw2BzR4.EbK_9AXnviLnLaw5SvIEJkyJo0dGtzUNCsEv4L8wXy3dH_YRDa
  IavIwk208zfgo6DGTbXvbnKGtoFLyNq10jbq.mo_GxL.A_v8Jnun3AEzT3Co
  WVJUflum7mXYoTLJfzJEds6aHURixFps0Qv1FolH.GVwHig2bP10xQS8ebjz
  P0F22tbs6cMyP9mfeE7L5.DpFRkkLT4y44wtQzXEJP.TBQx1dZBhG7jvXIiW
  o_FKijsUktrJL.p03zECvhBMhD8DSilWrKf19y0mxEcOENNq5HCpJVPPYEcm
  pRwOW0MnetIebnbhyzTV0zh_6Ghl72Kd8Ca_FywptMCyW8m7TGwZj01jws3
  FVI5WgNGF7gsWo9peDCSBWg_Jga.QAGUib7edzcb0ICdfbsmkJ78wSDxqyqA
  XyZJIKJ3ZioZt4pLAqWC0q5CrktJT2wzyltoaxvn8lbU1NQ_76VBnLEefQsB
  K60eEhu4xnUFSK6BhIMFlvrlsJ2ydX.C2dURnXH7mGGBwGDuTvhdX_GPI4Ys
  .k0a0qoU2KEEn59DhJqUAGETpvHhJvIyxVcbiUM.ZiNTv5cp3lwefZXnIQVuv
  D2PEVe9X0HJQetIZQMT2GLBmEqoY27fjtFcSWU_go99f2U2dDAahpALa7sac
  mU.kXpWfYCuL4HRW5Bi0upaJvQxxAfUnmHz0Uymx24yG_6EwqN0kxZQy6g0v
  qjXTvGnBhec_KmEqv7g2a.GA7D.2uRDTEfpXbI_I5w2tYoJtITUyVxosEJpr
  pyVIO.tPQmAGXaMQvqr05ZLc4uLcMt0YcW0oKNhN8FLoXpzzrB8EZI5H.9A1x
  vpV_C4e.0qHFg4vETxu_i_yl6mm860YxkLTyatkunIDy9HulJwrG0F7bm7DP
  uXjsmGo_.lN923Rg9uS0lxJAuCLc3jIVuVS_wgQ6JgYhymMoM_6y.d010DG
  vkRPM93iuFW9iufqFLqMG0nwQs8_9nSMKZfxNH3yd.2RMwiVtB.45G7aKd1D
  1h9z0VIy5Z2VldEGwmtmSKRtkSJNvUfLh0wVcebCs5m4NNXc.3P.xLdIwMk
  ND0u1ERVPNCmz0vzPTaMve0LR6ToPdISqEYF1RH6A0xUfkbM0Ewdcw1CaTf9
```

- From: Home Depot
- Sender's email address: [support@teckbe.com](mailto:support@teckbe.com)

## Defang ip address: 103.234.236.83

Defang IP Addresses

103.234.236.83

103[.]234[.]236[.]83

Which reputable organization is being spoofed in this phishing attempt?

Home Depot

✓ Correct Answer

What is the sender's email address?

support@teckbe.com

✓ Correct Answer

Inspect the email message source.  
What is the [defanged \(opens in new tab\)](#) X-Originating-IP ?

103[.]234[.]236[.]83

✓ Correct Answer

Continue analyzing the email message source.  
Which mail server generated the [Authentication-Results](#) header?

atlas102.free.mail.gq1.yahoo.com

✓ Correct Answer

## Task 7 Conclusion

In this room, you explored the anatomy of an email, starting with the structure of an email address and the path a message takes from sender to recipient. You then developed the technical skills needed to investigate suspicious emails by extracting and analyzing both the header and body source code. Finally, you examined common phishing techniques and applied your knowledge to analyze real-world examples, identifying attacker intent. Nicely done!

Before moving on, it's also important to understand Business Email Compromise (BEC), a type of attack where an adversary gains access to a legitimate internal email account and uses it to trick others into performing unauthorized or fraudulent actions.

### Further Learning

This room is the first of a series of rooms that make up the [Phishing Analysis](#) module of the [SOC Level 1 Path](#). You are encouraged to continue your learning with the walkthrough and challenge rooms below:

- [Phishing Emails in Action](#)
- [Phishing Analysis Tools](#)
- [Phishing Prevention](#)
- [The Greenholt Phish](#)
- [Snapped Phish-ing Line](#)

### Answer the questions below

What attack, signified by the acronym BEC, uses a compromised email to trick employees into fraud?

Business Email Compromise



✓ Correct Answer